

Project2 README

Tomas Salaz Math/CS 375- Numerical Methods

for 'root.m' MATLAB file, Is the script for Project #2, just press run.

Summary or Graph:

-The Legend of the graph shows the function used in Newtons Method [$f(x)$, Blue Line], the Base Function [$g(x)$, Red Line], and the Point on $g(x)$ that is closest to $(1,0)$. The BLUE line represent the Newtons Method results, the limit as the function approaches 0 is the x value for $g(x)$ that is closest to $(1,0)$

*NOTE: Lines 63,64 comments about adding abs function for a better graphical visual.

-The Command Window shows that the method took 17 iterations to get within the allowed tolerance. Line 21 has testing cases showing that better initial guesses result in less iterations.

-Command Window also show the $g(x)$ and y values for the base function x^2 , so the closes point to $(1,0)$ and the number of iterations if the initial value is changed